FUSION ELECTRONICS

Full Site Audit Report

Stack: React 18 · Node.js/Express · Material UI v5 · JSON-based data store

Audit Date: April 2, 2026

Prepared by: Chioma Ogbonna

Executive Summary

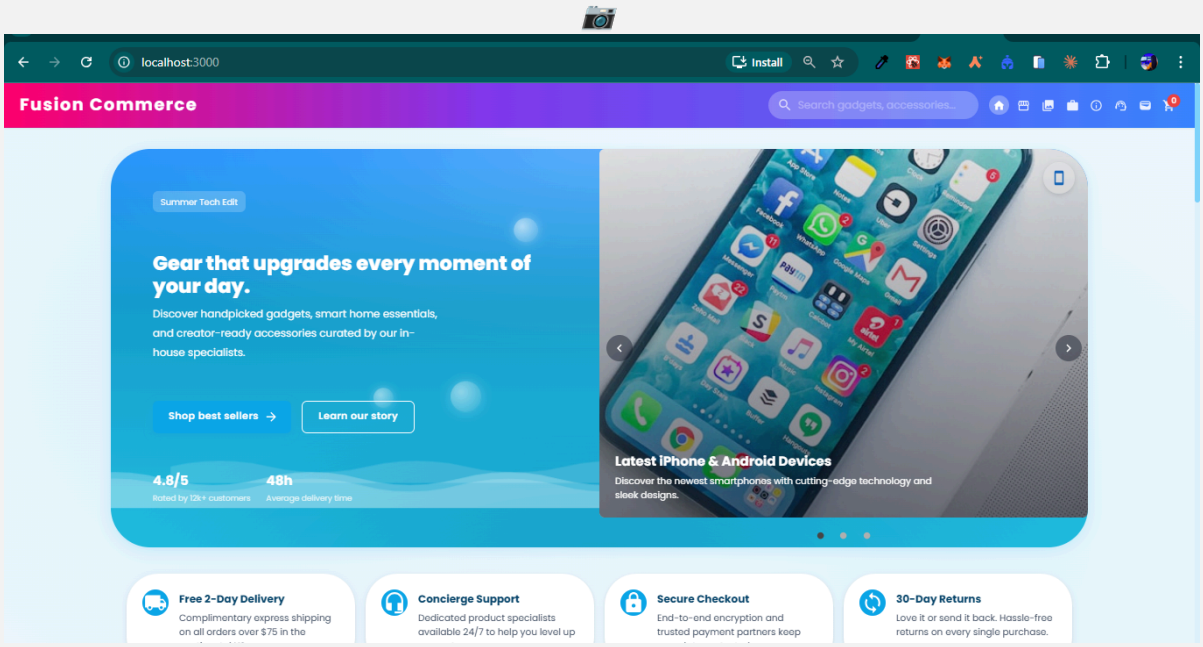
The application is a well-structured fullstack eCommerce site with a polished visual design. However, several critical functional bugs exist that would prevent real users from completing purchases, and a number of UX inconsistencies and missing features reduce the overall quality. The most severe issue is a broken checkout flow caused by a mismatch between the backend's data layer and the checkout route. Below is a page-by-page breakdown.

Priority Summary

Priority	Issue	Location
P0 — Critical	Checkout always fails (Product.find() stub)	*backend/routes/checkout.js*
P0 — Critical	Orders not persisted (lost on restart)	*backend/routes/checkout.js*
P0 — Critical	API base URL defaults to api.example.com	*src/services/apiClient.js*
P1 — High	Recommendations section always shows error	*Home.jsx, ProductDetails.jsx*
P1 — High	'Book Service' button non-functional	*src/pages/FreelancerServices.jsx*
P1 — High	Support contact form sends nothing	*src/pages/Support.jsx*
P1 — High	Newsletter form sends nothing	*Home.jsx, Footer.jsx*
P1 — High	No quantity support in cart/product	*Cart.jsx, ProductDetails.jsx*
P2 — Medium	Brand name inconsistency (mobile vs desktop)	*src/components/NavigationBar.jsx*
P2 — Medium	Duplicate category filters on Shop page	*src/pages/Shop.jsx*
P2 — Medium	Sort dropdown has no text label	*src/pages/Shop.jsx*
P2 — Medium	Add to Cart ignores out-of-stock status	*src/pages/ProductDetails.jsx*
P2 — Medium	Crypto mock exchange rates are hardcoded	*src/components/CryptoPaymentForm.jsx*

P2 — Medium	Error messages hidden with display:none	*Checkout.jsx, CheckoutForm.jsx*
P3 — Low	Cart badge shows 0 when empty	*src/components/NavigationBar.jsx*
P3 — Low	Price missing .toFixed(2) on detail page	*src/pages/ProductDetails.jsx*
P3 — Low	Social links are empty	*src/components/Footer.jsx*

Visual Audit — Home Page

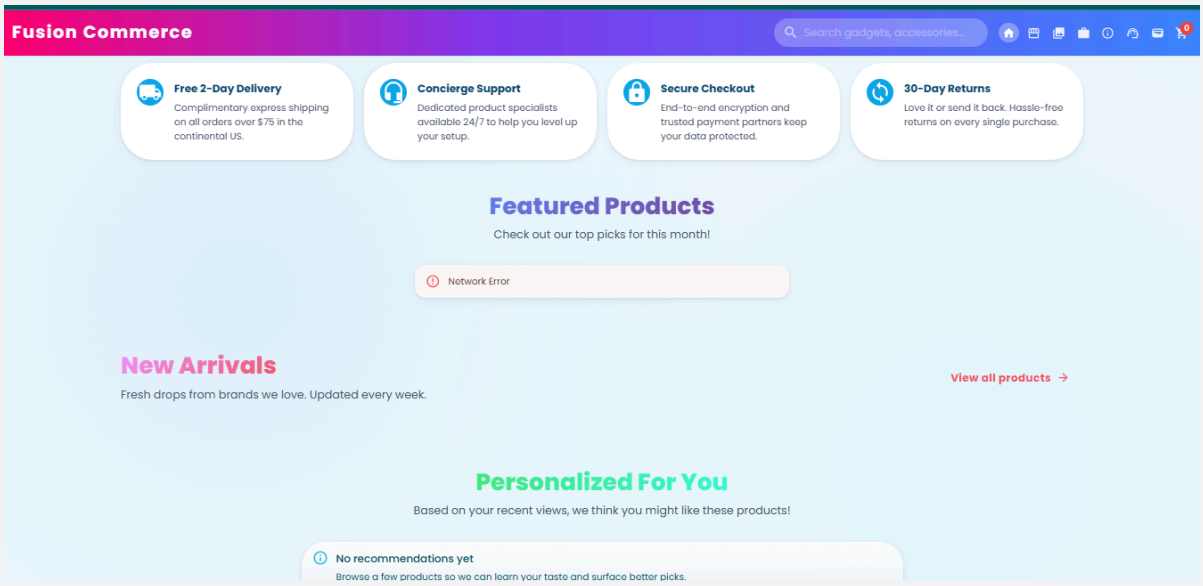


Screenshot: Home Page — Full View

[V1] Hero section is undersized and not full-width — The hero renders inside a constrained centered container rather than spanning the full viewport width. Should be edge-to-edge and visually commanding.

[V2] Navbar color scheme clashes with the page theme — The navbar uses a bold pink→purple→blue gradient while the rest of the page uses a calm light-blue/cyan palette. These two color systems fight each other.

[V3] Product grids are empty / not rendering — Both 'Featured Products' and 'New Arrivals' sections show no product cards. Either an API connectivity issue or a loading state that never resolves.



Empty Product Grid

[V4] Personalized For You error is visible to end users — A developer-facing Weaviate error message ('Weaviate cluster has been suspended') is prominently displayed in the middle of the home page.

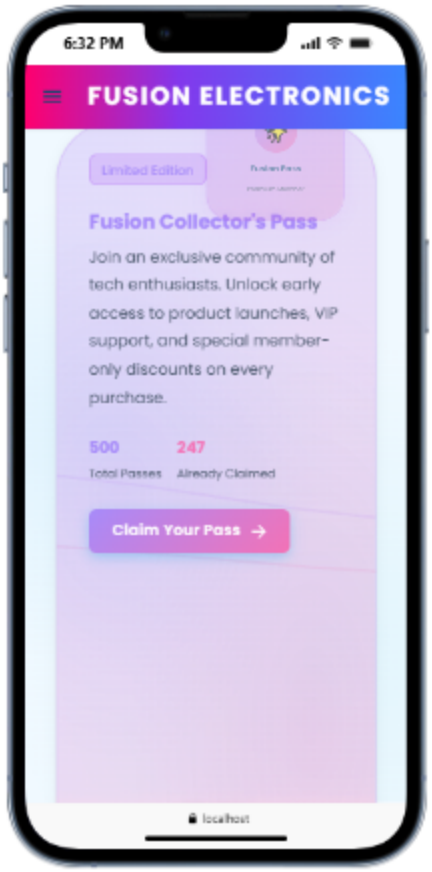
[V5] Section heading gradients are inconsistent — Each section uses a different gradient color for its heading text with no consistent design intent.

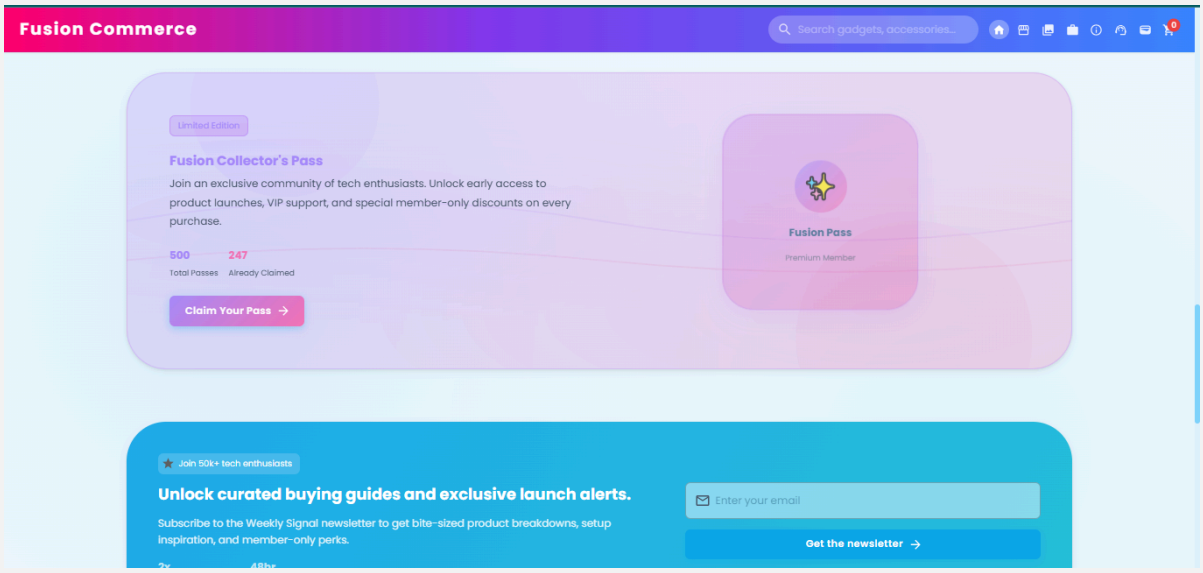
[V6] Page feels sparse due to empty product sections — Large empty gaps between sections due to product grids not populating. Skeleton loaders or placeholder cards needed.

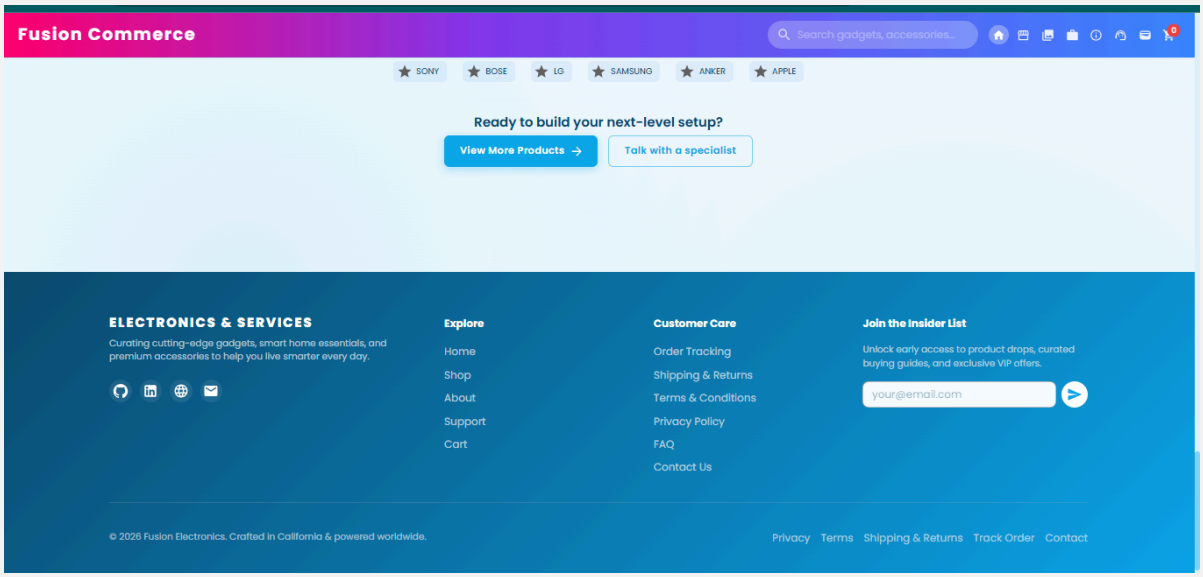
[V7] Value props row is well-executed — The four value proposition cards are clean, scannable, and well-spaced. This section works well as-is.

[V8] Hero carousel navigation arrows are barely visible — Carousel nav buttons use a semi-transparent dark color that blends into the hero's blue gradient background.

Home Page — Lower Half


6:32 PM
FUSION ELECTRONICS
Limited Edition
Fusion Collector's Pass
Join an exclusive community of tech enthusiasts. Unlock early access to product launches, VIP support, and special member-only discounts on every purchase.
500
Total Passes
247
Already Claimed
Claim Your Pass
localhost


Fusion Commerce
Search gadgets, accessories...
Limited Edition
Fusion Collector's Pass
Join an exclusive community of tech enthusiasts. Unlock early access to product launches, VIP support, and special member-only discounts on every purchase.
500
Total Passes
247
Already Claimed
Claim Your Pass
Fusion Pass
Premium Member
Join 50k+ tech enthusiasts
Unlock curated buying guides and exclusive launch alerts.
Subscribe to the Weekly Signal newsletter to get bite-sized product breakdowns, setup inspiration, and member-only perks.
Enter your email
Get the newsletter



📷 *Screenshot: Home Page — Lower Half*

[V9] Animated hero particles bleed into adjacent sections on mobile — z-index and overflow containment failure. The hero Paper component does not have overflow: hidden set. Fix: add overflow: hidden to the hero Paper wrapper.

[V10] Fusion Collector's Pass card is visually disconnected — Uses a pink/purple gradient introducing a third color system with no relationship to the site's established palette.

[V11] Newsletter section uses an unrelated dark teal background — Creates a jarring scroll experience immediately below the Collector's Pass card.

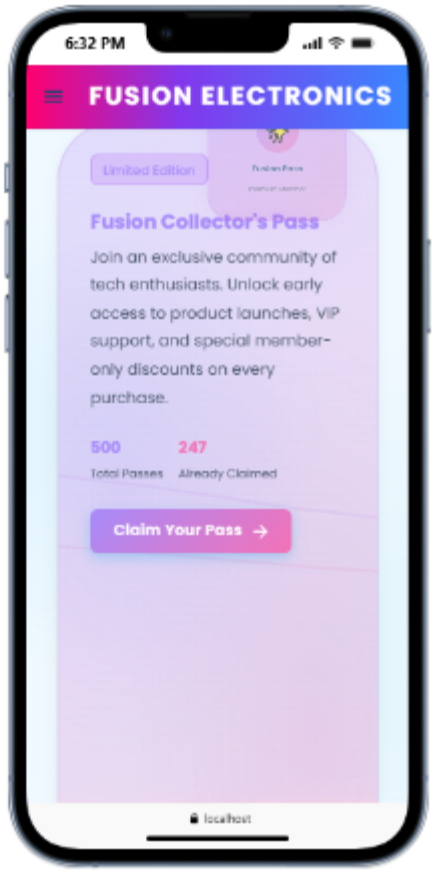
[V12] No visual rhythm or consistent section background treatment — The lower half cycles through completely unrelated background colors with no consistent pattern.

[V13] Brand logos row is plain text with no visual value — Six brand names rendered as plain text chips with no logos, hover states, or interactivity. 'ANKER' appears twice.

[V14] CTA floats in excessive whitespace — The bottom CTA section is surrounded by a disproportionate amount of empty white space above the footer.

[V15] Footer is the strongest section on the page — Well-structured with clear columns, newsletter input, social icons, and gradient background. Sets a good benchmark for the rest of the UI.

1. Navigation Bar



📷 *Screenshot: Navigation Bar — Mobile and Desktop*

[1.1] Brand name inconsistency — Mobile shows 'FUSION ELECTRONICS' but desktop shows 'Fusion Commerce'. Two different names for the same brand.

[1.2] Social links in footer are empty — All four social icon buttons have href='', navigating to the current page instead of external profiles.

[1.3] Search results dropdown position is fragile — Uses getBoundingClientRect() to position absolutely. On page scroll or window resize, the position becomes stale.

[1.4] Search shows all products when query is empty — Backend returns entire product catalog when query is empty or whitespace.

[1.5] No keyboard navigation in search results — Search results list is not keyboard-navigable. Users cannot arrow-key through results or dismiss with Escape.

[1.6] Cart icon always shows badge even at 0 — showZero is set on the cart badge, displaying '0' even when cart is empty. Visually noisy and non-standard.

2. Home Page (/)

[2.1] Personalized For You section is always broken — Depends on a Weaviate vector database explicitly noted as suspended in source code. Every visitor sees a warning alert.

[2.2] Featured Products only shows first 3 items — featuredProducts is hardcoded to products.slice(0, 3). No isFeatured flag is used — just the first three products.

[2.3] New Arrivals sort by createdAt but data has no real dates — Sort order is arbitrary as the JSON seed data has no meaningful createdAt values.

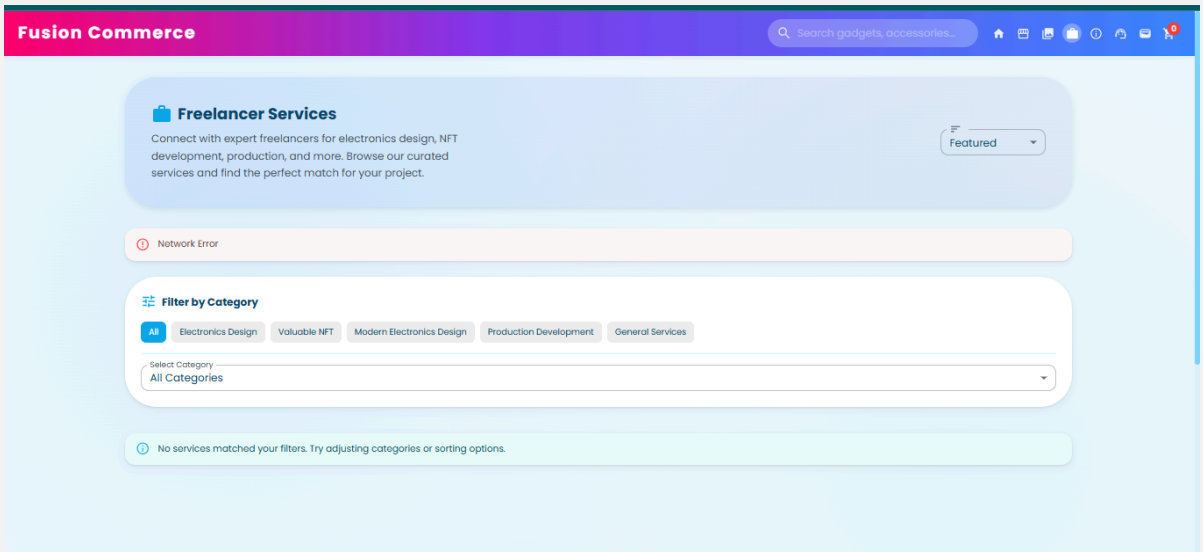
[2.4] Category cards don't apply filter correctly — URL param is ignored if categories haven't loaded yet due to useEffect dependency timing issue.

[2.5] Newsletter form submits but does nothing — Shows a success toast but makes no API call. No email is stored anywhere. Misleading to users.

[2.6] Hero carousel images load with no fallback dimensions — Carousel collapses to zero height before the fallback gradient kicks in, causing a layout shift.

[2.7] Animated bubble particles re-randomize on every render — Math.random() used for positioning directly in JSX — causes unnecessary re-renders.

3. Shop Page (/shop)



Screenshot: Shop Page — Filters and Product Grid

[3.1] Duplicate category filter controls — Two separate category filter mechanisms on the same page: Chip quick-filters AND a Select dropdown. Both control the same state.

[3.2] Sort label shows only an icon, no text — InputLabel for the sort Select contains only an icon with no text. Not accessible.

[3.3] URL category param only applied on mount — Filter may not apply correctly when navigating directly to /shop?category= before products load.

[3.4] In Stock filter has type dependency issue — Checks typeof product.stock === 'number'. If stock is a string from JSON, product is incorrectly excluded.

[3.5] No price range filter — No way to filter products by price range — a standard eCommerce feature.

[3.6] No search within the shop page — Users must use the global navbar search which navigates away from the shop.

[3.7] Loading spinner covers the entire viewport — Full-screen fixed overlay blocks all interaction. Skeleton loader would be more appropriate.

4. Product Detail Page

[4.1] Similar Products section is always broken — Frontend shows a 'Database cluster has been suspended' error with a placeholder URL link.

[4.2] Rating submission updates local state incorrectly — Frontend's optimistic update and backend's calculation use different formulas — displayed rating diverges on reload.

[4.3] Rating can be submitted multiple times — No check prevents a user from rating the same product multiple times, skewing the average.

[4.4] Product image has no loading=lazy — Main product image uses a plain img tag with no lazy loading attribute.

[4.5] Price displayed without currency symbol formatting — Price rendered without .toFixed(2), so $29.9 displays instead of $29.90.

[4.6] No quantity selector — Users can only add one unit of a product to the cart. No quantity input on the product detail page.

[4.7] Add to Cart does not check stock availability — Add to Cart button is always enabled even when product.stock === 0.

5. Cart Page

[5.1] No quantity adjustment in cart — Cart items can only be removed, not have their quantity changed. No way to order 2 of the same item.

[5.2] Cart total does not reflect quantity — calculateTotal() sums item.price directly — architecturally incorrect and will break if quantity support is added.

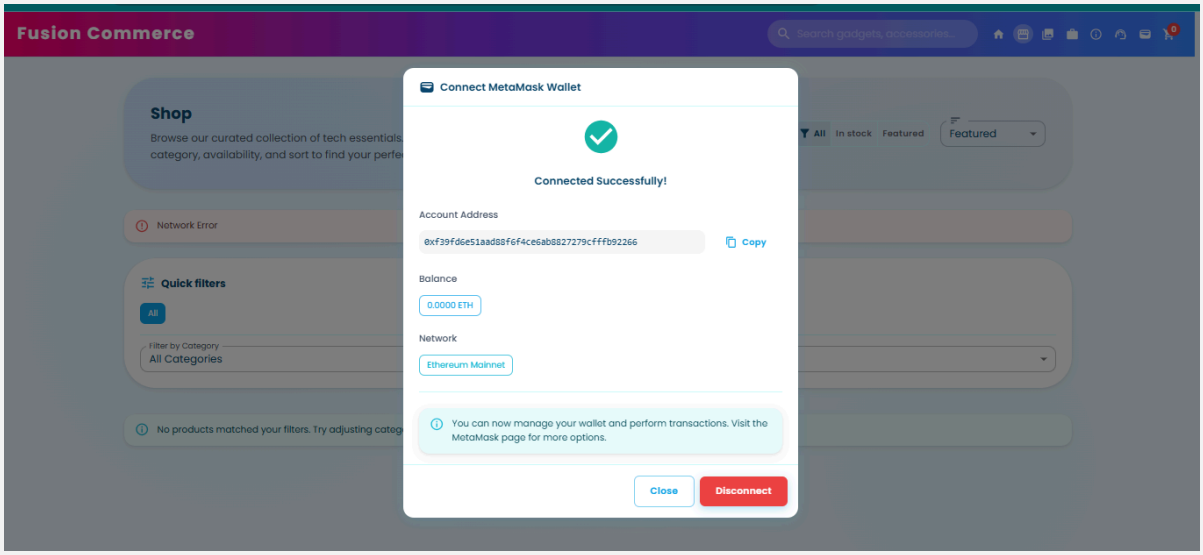
[5.3] Item price shown without currency symbol — Price shown as a plain number string with no $ prefix.

[5.4] No Clear Cart button — No way to empty the entire cart at once. Users must remove items one by one.

[5.5] Cart persists stale items across sessions — If a product is removed from the backend JSON, stale items remain if the API is down.

[5.6] Empty cart state CTA is generic — Generic empty cart message. Showing 2-3 featured products inline would reduce bounce.

6. Checkout Page (/checkout)



📷 *Screenshot: Checkout Page*

⚠️ **CRITICAL:** *This section contains the single most impactful bug in the entire application.*

[6.1] CRITICAL: Checkout always fails in production — The checkout route calls Product.find() which is a stub that always returns []. Every order submission fails with 'One or more products are no longer available.'

[6.2] Error message is hidden with display: none — Error messages use sx={{ display: 'none' }}. If a user misses the toast, they have no persistent indication of what went wrong.

[6.3] Checkout accessible with empty cart — Navigating directly to /checkout with an empty cart renders the form with 0 items and $0.00 total. No redirect.

[6.4] Shipping address is a single text field — No separate fields for street, city, state, ZIP, or country. Makes address validation impossible.

[6.5] Card validation inconsistency between frontend and backend — Frontend uses Luhn algorithm; backend uses a simple regex. A 12-digit number can pass the backend but fail Luhn.

[6.6] Crypto payment uses hardcoded mock exchange rates — BTC = $45,000 and ETH = $2,500 are hardcoded constants, not fetched from a live API. Always wrong.

[6.7] Mock wallet address regenerates on every render — Generated with Math.random() directly in the component body — changes while the user is trying to copy it.

[6.8] Item selection checkboxes add complexity without clear value — Allowing users to deselect cart items at checkout is an unusual pattern that adds cognitive load.

7. Order Success Page

[7.1] Page accessible without completing an order — Navigating directly to /order-success renders the success page with no order number, no email, and no items.

[7.2] Track order fails after server restart — Orders stored in-memory are lost on server restart, making all tracking lookups return 'order not found'.

8. Order Tracking Page (/order-tracking)

[8.1] Orders are stored in-memory and lost on restart — The existingOrders array is in-memory. Order.save() is a stub that does nothing. No data persistence.

[8.2] Support contact info uses placeholder values — support@example.com and +1 (833) 555-0195 are placeholders that need to be replaced with real contact info.

[8.3] Order tracking pre-fills stale data from localStorage — If the user has tracked a different order previously, their old order number and email appear in the form.

9. NFT Gallery (/nft-gallery)

[9.1] No pagination — All NFTs render in a single grid with no pagination or infinite scroll.

[9.2] Sort and filter params may not be implemented on backend — Frontend appends ?sort=...&rarity=... to the NFT API request but backend implementation is unconfirmed.

[9.3] No empty state illustration — Plain Alert with text. A more engaging empty state with illustration or CTA would improve the experience.

10. Freelancer Services (/freelancer-services)

[10.1] Categories are hardcoded in frontend — CATEGORIES array hardcoded in FreelancerServices.jsx. New backend categories won't reflect without a code change.

[10.2] Book Service button does nothing — handleBookService only shows a toast. Does not add to cart, navigate to checkout, or call any API. Entirely non-functional.

[10.3] No view filter toggle — Unlike the Shop page, no 'In Stock / Featured / All' toggle filter.

11. Static Pages

[11.1] Support contact form calls no API — Shows a success toast but makes no API call. No message is sent anywhere.

[11.2] About page milestones reference future dates as past — Fictional 2018-2024 milestones should be replaced with real content before client delivery.

[11.3] Footer social links are all empty — All four social icon buttons have href=''. Clicking them navigates to the current page.

12. Global / Cross-Cutting Issues

[12.1] No authentication or user accounts — No login, registration, or user session system. JWT_SECRET and MONGO_URI in .env.example suggest these were planned but not implemented.

[12.2] MongoDB URI in .env.example but app uses JSON files — Architecture mismatch is the root cause of the critical checkout bug.

[12.3] API base URL falls back to api.example.com — Without a proper .env file, the app fails to load any data.

[12.4] No .env file provided — Only .env.example exists. A new developer must know to create .env manually. Not documented in README.

[12.5] Rating updates are not persisted — PUT /api/products/:id/rating updates the in-memory array only. All ratings revert on server restart.

[12.6] No loading skeleton components — Full-page spinner or nothing while loading. Skeleton loaders would significantly improve perceived performance.

[12.7] No image optimization or lazy loading strategy — No width/height hints, no srcset, and inconsistent use of loading=lazy. Causes significant CLS and slow loads.

[12.8] console.warn/error calls left in production code — Multiple components log warnings and errors to the console. Should be removed before client delivery.

Recommended Next Steps

- **Fix the checkout data layer — Replace Product.find() stub with a direct lookup from the JSON file, or connect a real database.**
- Persist orders — Write orders to a JSON file or connect MongoDB so tracking works across restarts.
- Document environment setup — Add a SETUP.md or update README.md with required .env variables.
- Remove or replace the Weaviate dependency — Remove the recommendations sections or replace with the working Jaccard similarity fallback.
- Add quantity support — Implement quantity increment/decrement in both the cart and product detail pages.
- Fix all non-functional forms — Newsletter, support contact, and 'Book Service' should call real APIs or be clearly marked 'coming soon'.
- Replace placeholder content — Social links, support email, phone number, and company milestones should use real data.

— End of Report —